

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\Delta_{\pm} &= \frac{c}{24} \pm (I_H^{\pm} - 2I_S^{\pm} - I_C^{\pm} - I_W^{\pm}) + \frac{1}{2\pi} \Sigma_{\pm} \\ &\mp \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{dx}{2\pi} \varphi'_{\pm}(x) \mathcal{Q}_{\pm}(x),\end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}&\lfloor \frac{-2mp^i}{q-1} \rfloor + \lfloor \frac{mdp^i}{q-1} \rfloor + \lfloor \frac{-m(d-1)p^i}{q-1} \rfloor - \lfloor \frac{-mp^i}{q-1} \rfloor + 1 \\ &= \sum_{h=1}^{d-2} \lfloor \langle \frac{hp^i}{d-1} \rangle - \frac{mp^i}{q-1} \rfloor + \lfloor \langle \frac{p^i}{2} \rangle - \frac{mp^i}{q-1} \rfloor + \sum_{h=1}^{d-1} \lfloor \langle \frac{-hp^i}{d} \rangle + \frac{mp^i}{q-1} \rfloor.\end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\mathbf{E}(\mathbf{x}, t) &= \frac{1}{4\pi} \int \rho_e(\mathbf{x}', t-r) \frac{\mathbf{r}}{r^3} d^3x' \\ &+ \frac{1}{4\pi} \int \mathbf{J}_m(\mathbf{x}', t-r) \times \frac{\mathbf{r}}{r^3} d^3x' - \frac{1}{4\pi} \int \frac{1}{r} \frac{\partial \mathbf{J}_m(\mathbf{x}', t-r)}{\partial t} d^3x',\end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\Gamma^5 = \begin{pmatrix} 0 & 1_2 \\ 1_2 & 0 \end{pmatrix}; \quad \Gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}; \quad \Gamma^0 = \begin{pmatrix} 1_2 & 0 \\ 0 & -1_2 \end{pmatrix}; \quad \Gamma^4 = -i\Gamma^5 = \begin{pmatrix} 0 & i_2 \\ i_2 & 0 \end{pmatrix}.$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\bar{\omega}^m \left(\frac{b^{d-1} d^d}{a^d (d-1)^{d-1}} \right) \prod_{i=0}^{r-1} \prod_{h=1}^{d-2} \frac{\Gamma_p(\langle (\frac{h}{d-1} - \frac{m}{q-1}) p^i \rangle)}{\Gamma_p(\langle \frac{hp^i}{d-1} \rangle)} \prod_{\substack{h=1 \\ h \neq \frac{d}{2}}}^{d-1} \frac{\Gamma_p(\langle (\frac{h}{d} + \frac{m}{q-1}) p^i \rangle)}{\Gamma_p(\langle \frac{hp^i}{d} \rangle)} = 1.$$